2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer:(penalty regime: 0 %)

Feedback

#include <stdio.h>

```
int main()
{
   int n;
   scanf("%d", &n);

   int arr[n];

   for (int i = 0; i < n; i++)
      scanf("%d", &arr[i]);

   int slow = arr[0];
   int fast = arr[0];

   // Find intersection point
   do
   {
      slow = arr[slow];
      fast = arr[arr[fast]];
   } while (slow != fast);

   // Find duplicate number
   slow = arr[0];

   while (slow != fast)
   {
      slow = arr[slow];
      fast = arr[fast];
```

```
    }

    printf("%d", slow);

    return 0;
}
```

	Input	Expected	Got	
	11 10 9 7 6 5 1 2 3 8 4 7	7	7	
	5 1 2 3 4 4	4	4	
	5 1 1 2 3 4	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.